

**COMP 410**  
**Fall 2026**  
**Midterm Practice Exam #1**

This is representative of the kinds of topics and kind of questions you may be asked on the midterm. This practice exam, along with assignments 1-2 and the first 4 in-class worksheets (everything through Recursive Procedures), are intended to be comprehensive of everything on the exam. That is, I will not ask anything that's not somehow covered by those sources.

You are permitted to bring two 8.5 x 11 sheets of paper into the exam with you, and these may have anything on them, either printed or handwritten. Both sides of both sheets can be used.

**Abstract Syntax Trees**

In Boolean expressions,  $\neg$  has the highest precedence, followed by  $\wedge$  and  $\vee$ . With this in mind, write out the ASTs corresponding to each of the following Boolean expressions:

1.)  $\neg a \wedge b \vee c$

2.)  $(a \vee b) \wedge c$

3.)  $\neg (a \wedge b) \wedge (b \vee c)$

Arithmetic expressions can be used to form Boolean expressions with the help of arithmetic comparisons (e.g.,  $<$ ,  $\leq$ ,  $>$ ,  $\geq$ ,  $==$ ). These comparisons have the lowest possible precedence. With this in mind, write out the ASTs corresponding to each of the following expressions:

4.)  $1 * 2 + 3 == 4$

$$5.) (2 + 2 < 4) \wedge \neg a$$

6.) Consider the following Python class definitions, which are adapted from assignment 1's boolean evaluator. These classes are used to represent AST nodes.

```
class And:
    def __init__(self, left, right):
        self.left = left
        self.right = right

class Or:
    def __init__(self, left, right):
        self.left = left
        self.right = right
```

Assume that Boolean true is represented as an AST with Python's `True`, and Boolean false is represented as an AST with Python's `False`. With all this in mind, represent the following Boolean expressions in Python using `And`, `Or`, `True`, and `False` as appropriate.

6.a) `true ∧ false`

6.b.) `false ∨ true`

6.c.) `false ∧ true ∨ true`

6.d.) `false ∨ true ∧ true`

## Semantic Tableau

For each of the following Boolean formulas, write out the complete semantic tableau tree. **Circle** the nodes in the tree representing solutions. If a tree has no solutions, say so. **Be sure to write all steps.**

7.)  $\neg a \wedge a$

8.)  $(a \vee \neg a) \wedge a$

9.)  $(\neg x \wedge \neg y) \vee (x \wedge y)$

## Prolog - Modeling the World

10.a)

For this problem, you need to write a clause database encapsulating pricing information for a convenience store. Write Prolog code accurately reflecting the following:

- Soda costs \$2
- Chips cost \$3
- Hot dogs cost twice as much as soda (do not hardcode \$4)
- Soda chips, and hot dogs are food
- Pencils and pens are office supplies
- All office supplies cost \$2
- Cold medicine costs \$7

Using the clause database you previously wrote, write queries to determine the following:

10.b.) Which items cost exactly \$2?

10.c.) Which items cost more than \$3?

10.d.) Which foods cost less than \$3?

10.e.) Which foods are also office supplies?

## Unification

Consider each of the following unification attempts. If the unification succeeds, record any values any variables take. If the unification fails, say so.

11.)  $\text{foo}(1, X) = \text{foo}(Y, 2)$

12.)  $\text{foo}(1, X) = \text{foo}(X, 2)$

13.)  $\text{foo}(1, \_) = \text{foo}(X, 2)$

14.)  $\text{foo}(1, \_) = \text{foo}(1, \_)$

15.)  $\text{foo}(1, 2, \text{bar}) = \text{foo}(X, \_, \_, \_)$

16.)  $\text{foo}(\text{bar}(\text{baz}), X) = \text{foo}(Y, Z), Y = \text{bar}(A)$